Paso 1: Instalar Apache
Se usa “sudo apt install apache2” ya que es en Ubuntu

Paso 2: Activar SSL en apache
se usa “sudo a2enmod ssl””

```
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
```

Paso 3: Crear certificado autofirmado

```
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:CR
State or Province Name (full name) [Some-State]:San Jose
Locality Name (eg, city) []:San Pedro
Organization Name (eg, company) [Internet Widgits Pty Ltd]:UCR
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:Agustin
Email Address []:
```

Algunos campos del certificado fueron omitidos debido a que no son obligatorios para el funcionamiento del certificado autofirmado en este contexto de prueba.

Paso 4: Configurar Apache para usar HTTPS

```
#   A self-signed (snakeoil) certificate can be created by installing
#   the ssl-cert package. See
#   /usr/share/doc/apache2/README.Debian.gz for more info.
#   If both key and certificate are stored in the same file, only the
#   SSLCertificateFile directive is needed.
SSLCertificateFile      /etc/ssl/certs/apache-selfsigned.crt
SSLCertificateKeyFile   /etc/ssl/private/apache-selfsigned.key

#   Server Certificate Chain:
#   Point SSLCertificateChainFile at a file containing the
#   concatenation of PEM encoded CA certificates which form the
#   certificate chain for the server certificate. Alternatively
#   the referenced file can be the same as SSLCertificateFile
#   when the CA certificates are directly appended to the server
```

Paso 5: Activar el sitio SSL y probarlo en el Navegador
Primero “sudo a2ensite default-ssl”

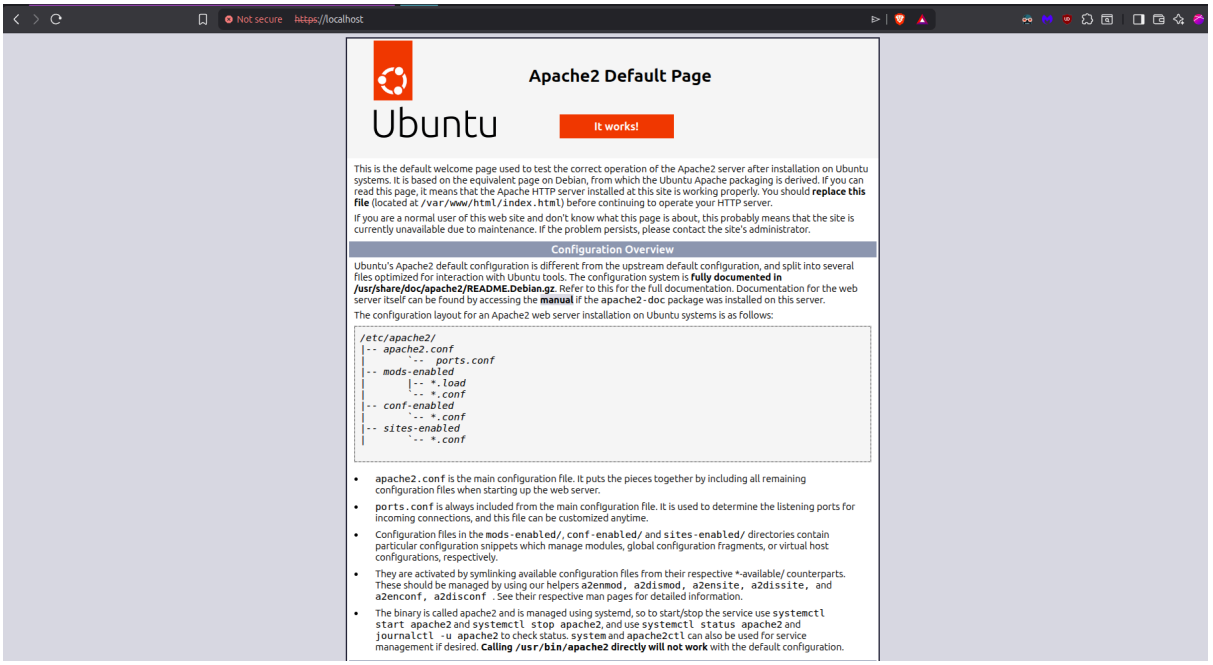
```
Enabling site default-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
```

Luego “sudo systemctl restart apache2” y verificar que está activo con “sudo systemctl status apache2”

```
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-04-21 08:10:28 CST; 1min 31s ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 8746 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 8749 (apache2)
      Tasks: 55 (limit: 18232)
     Memory: 6.5M (peak: 8.6M)
        CPU: 72ms
     CGroup: /system.slice/apache2.service
             ├─8749 /usr/sbin/apache2 -k start
             ├─8751 /usr/sbin/apache2 -k start
             └─8752 /usr/sbin/apache2 -k start

abr 21 08:10:28 agustins-ubuntuhp systemd[1]: Starting apache2.service - The Apache HTTP Server...
abr 21 08:10:28 agustins-ubuntuhp apachectl[8748]: AH00558: apache2: Could not reliably determine the server's fully quali
abr 21 08:10:28 agustins-ubuntuhp systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
```

Ahora se pruebe en el navegador y ver si funciona:



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|       `--  ports.conf
|-- mods-enabled
|       |-- *.load
|       `-- *.conf
|-- conf-enabled
|       `-- *.conf
|-- sites-enabled
|       `-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective *-available/ counterparts. These should be managed by using our helpers `a2enmod, a2dismod, a2ensite, a2dissite,` and `a2enconf, a2disconf` . See their respective man pages for detailed information.
- The binary is called apache2 and is managed using systemd, so to start/stop the service use `systemctl start apache2` and `systemctl stop apache2`, and use `systemctl status apache2` and `journalctl -u apache2` to check status. `system` and `apache2ctl` can also be used for service management if desired. **Calling `/usr/bin/apache2` directly will not work** with the default configuration.

Document Roots

By default, Ubuntu does not allow access through the web browser to *any* file outside of those located in `/var/www`, **public_html** directories (when enabled) and `/usr/share` (for web applications). If your site is using a web document root located elsewhere (such as in `/srv`) you may need to whitelist your document root directory in `/etc/apache2/apache2.conf`.

The default Ubuntu document root is `/var/www/html`. You can make your own virtual hosts under /var/www.

Reporting Problems

Please use the `ubuntu-bug` tool to report bugs in the Apache2 package with Ubuntu. However, check **existing bug reports** before reporting a new bug.

Please report bugs specific to modules (such as PHP and others) to their respective packages, not to the web server itself.

Referencias

guide, s., Yadav, U. K., & Naphad, S. (2023, November 28). *Configuring a Self-Signed SSL Certificate on a Apache Server*. Medium. https://medium.com/@pasanglamatamang/configuring-a-self-signed-ssl-certificate-on-a-apache-server-cbcd6eefdf1a

Vidrio, E., & Jamón, C. (2022, April 25). *How To Create a Self-Signed SSL Certificate for Apache in Ubuntu 22.04*. Digital Ocean. https://www.digitalocean.com/community/tutorials/how-to-create-a-self-signed-ssl-certificate-for-apache-in-ubuntu-22-04